DAY 21 | HANDS-ON ACTIVITIES

Azure Deployment and Continuous Delivery: Problem Statements

Seven problems. Every problem runs offline in the notebook. Steps that create Azure resources are marked optional; do them in class only, in the subscription window, and delete them afterwards.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Choose the Host

Level: Warm-up | **Concepts:** App Service, Container Apps, AKS, VMs

For six workloads (a FastAPI API, a nightly batch job, a legacy Windows app, a 40-service platform, a static React site, an event-driven worker) choose a host and give a one-line reason.

Sample run / expected result

```
FastAPI API      -> App Service or Container Apps: simple PaaS
nightly batch    -> Container Apps job: runs and stops
legacy Windows   -> VM: needs a full OS
```

Hints

- Start with the simplest option that meets the need.

Problem 2: CD Pipeline YAML

Level: Medium | **Concepts:** stages, environments, approvals

Write a pipeline with Test, Build (Docker@2 buildAndPush to ACR, tag with Build.BuildId), DeployDev and DeployProd (deployment jobs targeting dev and production environments). Validate it with yaml.safe_load and print the stage order.

Sample run / expected result

```
Test -> Build -> DeployDev -> DeployProd
DeployProd environment: production
```

Hints

- Approvals are configured on the environment in the Azure DevOps UI, not in YAML.

Problem 3: Variables or Secrets?

Level: Easy | **Concepts:** pipeline variables, masking

Sort twelve settings (region, image tag, DB password, JWT secret, app name, API key, log level, storage connection string and others) into plain variables and Key Vault secrets. Write a mask() function that hides secret values in a log line.

Sample run / expected result

```
plain : region, imageTag, appName, logLevel ...
secret: dbPassword, jwtSecret, apiKey ...
log   : connecting with password=***
```

Hints

- If it would hurt to see it in a screenshot, it is a secret.

Problem 4: Key Vault with a Managed Identity NEW (v4)

Level: Medium | **Concepts:** Key Vault, managed identity, least privilege

Using the offline vault simulator in the notebook, give the app an identity, grant it only the Key Vault Secrets User role, and read db-url. Show that an identity without the role is refused, and that a rotated secret is picked up without a code change.

Sample run / expected result

```
tasks-api (Secrets User) -> db-url v1 OK
report-job (no role)     -> 403 Forbidden
after rotation           -> db-url v2 OK
```

Hints

- The simulator mirrors the real flow: identity, role, token, read.

Problem 5: Spot the Leak NEW (v4)

Level: Easy | **Concepts:** secrets hygiene

Review a pipeline YAML and a Dockerfile that contain four secret mistakes (password in YAML, echo of a secret, ENV with a key, connection string in appsettings). Fix each with Key Vault, managed identity or secret variables.

Sample run / expected result

```
Line 12: DB_PASSWORD in YAML -> Key Vault reference
Line 20: echo $(apiKey)     -> remove
```

Hints

- Search for password, secret, key, token and connection.

Problem 6: Release Simulator

Level: Challenging | **Concepts:** dev, approval, prod, smoke test, rollback

Use the notebook's release simulator to promote image tags 101 and 102. Make tag 102 fail its smoke test in dev so it never reaches prod. Then force a bad tag into prod and roll back to the previous tag.

Sample run / expected result

```
101: dev OK -> approved -> prod OK
102: dev smoke test FAIL -> stopped
103: prod FAIL -> rolled back to 101
```

Hints

- Keep a history list per environment so rollback knows the previous good tag.

Problem 7: Runbook

Level: Stretch | **Concepts:** operations, rollback, rotation

Write a one-page runbook for SmartDesk: deploy, verify, read logs, roll back to the previous image tag, rotate a Key Vault secret, and clean up resources.

Sample run / expected result

```
Rollback: az containerapp update -n smartdesk-api -g rg-prod --image <acr>/smartdesk-api:<pre
```

Hints

- Write it for a teammate who has never seen the app.